

# **INDIVIDUAL TECHNICAL REPORT**

## **Predictive Analytics using Portal**

### **Environmental Problem: Water Wastage**

SDG Focus: Sustainable Development Goal 6  
Clean Water and Sanitation

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# ABSTRACT

Water is an essential resource, but inefficient consumption, leakage and excessive usage contribute to significant water wastage. Conventional monitoring methods mainly analyse past consumption and may identify problems only after wastage has occurred.

This report proposes a **Predictive Water Wastage Analytics Portal** that uses historical water-consumption data to predict future demand and identify abnormal consumption patterns. The system compares predicted and actual consumption and generates alerts when significant deviations occur. This can help users investigate possible leakage or inefficient usage at an early stage.

The proposed solution supports **SDG 6 – Clean Water and Sanitation**, particularly Target 6.4, which focuses on improving water-use efficiency. The report discusses the technical approach, sustainability, feasibility and expected environmental benefits of the proposed system.

## 1. INTRODUCTION

Water is a vital natural resource required for human life, agriculture, industries and economic activities. Increasing population and urbanisation have increased the demand for freshwater, making efficient water management increasingly important.

Water wastage can occur due to leakage, excessive consumption, overflowing tanks, inefficient fixtures and poor monitoring. Traditional monitoring methods generally record past consumption but provide limited information about future demand or unusual usage.

**Predictive analytics** provides a way to address this limitation by using historical data to identify patterns and predict future consumption. A predictive system can establish an expected level of water use and identify significant deviations from it.

This report proposes a **Predictive Water Wastage Analytics Portal** that uses historical consumption data to forecast future water demand and detect potentially abnormal consumption. The system is intended to support proactive water management and contribute to the objectives of **SDG 6 – Clean Water and Sanitation**.

## 2. PROBLEM STATEMENT

Water wastage is often detected only after excessive consumption has already occurred. Manual meter readings and conventional monitoring systems provide historical information but may not identify abnormal usage at an early stage.

For example, an unexpected increase in consumption may be caused by a leak, overflowing tank or inefficient usage. However, it may also result from legitimate factors such as increased occupancy or a special event. Therefore, a system is required that can analyse historical patterns and determine when current or future consumption differs significantly from the expected level.

### **Problem Definition**

*How can historical water-consumption data be analysed to predict future demand and identify abnormal consumption patterns that may indicate water wastage?*

The proposed Predictive Water Wastage Analytics Portal addresses this problem through forecasting and anomaly detection.

## 3. SUSTAINABILITY ANALYSIS

### **. 3.1 SDG 6 – Clean Water and Sanitation**

The proposed solution directly supports **Sustainable Development Goal 6 (SDG 6)**, which aims to ensure the availability and sustainable management of water and sanitation for all. The strongest connection

is with **Target 6.4**, which focuses on improving water-use efficiency and ensuring sustainable freshwater withdrawals.

The proposed portal contributes to this target by:

- predicting water demand,
- identifying abnormal consumption,
- supporting early detection of possible wastage,
- improving monitoring of water-use efficiency.

The basic sustainability cycle is:

**Historical Data → Prediction → Abnormality Detection → Corrective Action → Reduced Wastage → Improved Water Efficiency**

### **3.2 ISO 14001**

ISO 14001 provides a framework for environmental management. The proposed portal can support an organisation's environmental management by helping it monitor water consumption, establish reduction targets, identify abnormal usage and evaluate environmental performance.

However, using the portal alone does not make an organisation ISO 14001 certified.

### **3.3 LCA/EIA Consideration**

A full Environmental Impact Assessment is generally not required for a software-based solution. However, its environmental impacts should still be considered.

The portal requires electricity and computing resources, while optional smart meters may generate electronic waste. These impacts are expected to be relatively small compared with the potential benefit of reducing unnecessary water consumption.

## 4. PROPOSED ENGINEERING / IT SOLUTION

The proposed solution is a **Predictive Water Wastage Analytics Portal**, a web-based system that analyses historical water-consumption data and predicts future demand.

The system follows the process:

*Data Collection → Data Processing → Prediction → Comparison → Anomaly Detection → Alert → Corrective Action*

The portal can initially use historical meter readings or water bills. Additional information such as occupancy, weather, rainfall and seasonal patterns can be incorporated when available.

The main components are:

1. **Data Collection Module** – collects historical water-consumption records.
2. **Data Processing Module** – cleans and organises the collected data.
3. **Prediction Module** – forecasts future water demand.
4. **Anomaly Detection Module** – identifies significant deviations from expected consumption.
5. **Dashboard** – displays consumption, predictions and alerts.
6. **Alert and Recommendation Module** – informs users about unusual consumption and suggests possible areas for investigation.

## 5. PREDICTIVE ANALYTICS MODULE

The central component of the portal is the predictive model.

The model learns relationships between historical observations and future consumption.

Possible algorithms include:

- Linear Regression
- Multiple Linear Regression

- ARIMA
- Random Forest
- Gradient Boosting
- Artificial Neural Networks
- LSTM networks

For an initial conceptual implementation, a **time-series forecasting model or a suitable machine-learning regression model** can be selected based on the quantity and quality of available data.

Research has shown that machine-learning and time-series methods can be useful for water-demand forecasting, while the most appropriate model depends on the characteristics of the available data and application.

## 6.WATER WASTAGE DETECTION

Prediction alone cannot confirm water wastage, as increased consumption may have legitimate causes such as higher occupancy or special events. Therefore, the portal combines **forecasting with anomaly detection**.

If actual consumption consistently exceeds the predicted range, the system can flag it as a potential wastage event and recommend checking for leaks, overflowing tanks, faulty fixtures or excessive usage.

## 7.PORTAL FEATURES

The proposed portal provides the following features:

- **Dashboard:** Displays current, predicted and historical water consumption.

- **Prediction:** Forecasts future water demand using historical data.
- **Anomaly Detection:** Identifies unusually high consumption.
- **Alerts:** Notifies administrators when abnormal usage is detected.
- **Recommendations:** Suggests possible causes and actions to investigate.
- **Reports:** Provides daily, weekly and monthly consumption trends.

## 8.UNIQUE ASPECT OF THE PROPOSED SOLUTION

The unique aspect of the proposed solution is its **predictive approach**. Instead of simply displaying past water consumption, the portal establishes an expected consumption level and compares it with actual usage.

Thus, it changes water management from **reactive monitoring to proactive detection**, helping users identify possible wastage at an earlier stage.

## 9.FEASIBILITY ANALYSIS

### 9.1 Technical Feasibility

The system can be developed using commonly available technologies such as Python, machine-learning libraries, databases and web-development frameworks. Initially, historical meter readings can be used, while smart meters can be integrated in future versions.

### 9.2 Economic Feasibility

A basic version can use existing meter data and open-source software, keeping initial costs low. Reduced water consumption and earlier detection of leaks can potentially lower water and maintenance costs.

### **9.3 Operational Feasibility**

The portal can be operated through a simple dashboard. Administrators can view predictions, receive alerts and investigate abnormal consumption without requiring advanced technical knowledge.

### **9.4 Data Feasibility**

The system can use daily or monthly meter readings, with additional information such as occupancy, weather and seasonal patterns added when available. Prediction quality will depend on the accuracy and quantity of the available data.

## **10.EXPECTED ENVIRONMENTAL BENEFITS**

The proposed system is expected to provide the following benefits:

- Reduction in unnecessary water consumption.
- Early identification of possible leaks and abnormal usage.
- Improved water-use efficiency.
- Better planning of future water demand.
- Reduced energy associated with unnecessary water treatment and pumping.
- Increased awareness of water conservation.

These benefits support **SDG 6**, particularly the objective of improving water-use efficiency.

## 11.LIMITATIONS

The effectiveness of the system depends on the quality and availability of historical data. Sudden changes in occupancy, weather or user behaviour may affect prediction accuracy. False alerts may also occur because high consumption does not always indicate wastage.

Therefore, alerts should be treated as **indications requiring human verification**, rather than definite proof of water wastage.

## 12.JUSTIFICATION OF THE PROPOSED SOLUTION

Predictive analytics provides a more proactive approach to water management than conventional monitoring. Instead of identifying excessive consumption only after it occurs, the proposed portal predicts expected demand and highlights unusual deviations.

The solution is also scalable and can be adapted for residential buildings, educational institutions, hotels, offices and other facilities. Its ability to use existing historical data makes it suitable for gradual implementation.

Therefore, the proposed system provides a practical IT-based approach to improving water-use efficiency and supporting sustainable water management.

## 13. FUTURE ENHANCEMENTS

Future versions of the system can include:

- Real-time IoT/smart-meter integration.
- Automated leak detection.
- Weather-based prediction.
- Mobile application and notifications.
- Explainable AI for prediction results.
- Automated control of valves and irrigation systems.

These additions can make the system more responsive and effective.

## 14. CONCLUSION

Water wastage is a significant challenge that requires efficient monitoring and timely intervention. The proposed **Predictive Water Wastage Analytics Portal** uses historical consumption data to forecast future water demand and identify abnormal usage patterns.

By comparing predicted and actual consumption, the system can provide early warnings and support corrective action. The solution aligns with **SDG 6**, particularly the goal of improving water-use efficiency, and can also support environmental management practices.

As a conceptual proposal, its actual performance would need to be evaluated using real-world data. Nevertheless, predictive analytics offers a promising approach for shifting water management from **reactive monitoring to proactive conservation**.

# 15. REFERENCES

1. United Nations Department of Economic and Social Affairs. **Goal 6: Clean Water and Sanitation – Sustainable Development Goals**. UN SDG Knowledge Platform.
2. UN-Water. **Progress on Change in Water-Use Efficiency – 2024 Update**. UN-Water, 2024.
3. International Organization for Standardization (ISO). **ISO 14001:2026 – Environmental Management Systems – Requirements with Guidance for Use**. ISO, 2026.
4. U.S. Environmental Protection Agency. **Life Cycle Assessment: Principles and Practice**. EPA.
5. Ahmed, A. A., Sayed, S., Abdoulhalik, A., Moutari, S., & Oyedele, L. **Applications of machine learning to water resources management: A review of present status and future opportunities**. *Journal of Cleaner Production*, 2024, 140715.
6. House-Peters, L. A., & Chang, H. **Urban water demand modeling: Review of concepts, methods, and organizing principles**. *Water Resources Research*, 2011.
7. *Forecasting urban household water demand with statistical and machine learning methods using large space-time data: A comparative study*. *Environmental Modelling & Software*, 2018.
8. *Explainable artificial intelligence for reliable water demand forecasting to increase trust in predictions*. *Water Research*, 2025.
9. Koh, V. K. Z., et al. **A review of recent hybridized machine learning methodologies for time series forecasting on water-related variables**. *Journal of Hydrology*, 2025.